Banking instantiation: applying the framework to SASB FN-CB

six commercial-banks topics → 20 sustainability reports → per-report MMRI

Layer A Regulatory benchmark + corpus

SASB Commercial Banks Standard (FN-CB) — $F = \{f_1, \ldots, f_6\}$

Data Security	Financial Inclusion	ESG in Credit	Financed Emissions	Business Ethics	Systemic Risk
FN-CB-230a	FN-CB-240a	FN-CB-410a	FN-CB-410b	FN-CB-510a	FN-CB-550a

Corpus (this study)

Big Four AU banks: ANZ, CBA, NAB, Westpac
2021 – 2025 → 20 reports

benchmark + reports

Layer B Computational assessment

B1 Extract & retrieve

PDF text extraction (pdfplumber)
→ ~150-word passages
→ top-K passages per topic f_i

B2 Score coverage (LLM)

Claude | Gemini structured prompt

$C_{r,i} = \alpha\,\text{Pres} + \beta\,\text{Spec} + \gamma\,\text{Quant} \in [0,1]$

$\alpha + \beta + \gamma = 1$

B3 Per-topic misalignment

$M_{r,i} = 1 - C_{r,i} \in [0,1]$

0 = fully disclosed
1 = silent on a required topic

Materiality Misalignment Risk Index (MMRI)

$$\text{MMRI}_r = \frac{\sum_i a_i w_i M_{r,i}}{\sum_i a_i w_i} \in [0,1] \qquad a_i \in \{0,1\} \text{ applicability}, \quad \sum_i w_i = 1$$

this study: $a_i = 1\ \forall i$, $w_i = 1/6$ (equal-weight baseline)

reported as $\overline{\text{MMRI}}_r \pm$ 95% CI over k runs

Validation

human-annotated
reference KG

ablation studies
+ repeated k runs

comparable risk signal → decision-oriented outputs

Layer C Decision-oriented outputs

Per-report heatmap

$M_{r,i}$ across all reports
(bank × year × topic)
[fig 1]

Topic-level bars

5-yr mean misalignment
by bank × topic
[fig 2]

MMRI trend

MMRI time-series
per bank, 2021 – 2025
[fig 3]

supports: Investors (disclosure-risk complement to ESG ratings) · Firms (topic-level reporting gaps) · Regulators (scalable alignment monitoring)

solid arrows = data flow dashed arrows = validation feedback